Free-Surface RANS Simulation of a 1 m Boat Model on a 300,000-Cell Budget: Mesh Design and the Resistance Penalty of a Bow Plate

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Individual term project in the Smart Computational Fluid Dynamics course (report submitted 15 December 2025), revised and condensed by the author in November 2026.

Abstract. The calm-water resistance of a 1 m boat model at 1.21 m/s (Froude number 0.39) was computed with an unsteady RANS solver, the k–ω SST turbulence model and a volume-of-fluid free surface, under a limit of about 300,000 cells. Because the transverse wavelength at this speed (0.94 m) is almost equal to the hull length, the domain was kept at 3.5 L ahead of the bow and 5.5 L behind the stern, and the remaining cells were distributed among free-surface, wake, hull and bow/stern refinement zones. A first attempt to resolve the viscous sublayer ($y+ \approx 1$) placed fifteen prism layers inside a stack covering only 19 % of the boundary-layer thickness; a wall-function approach with $y+ \geq 30$ was adopted instead because a consistent wall-resolved mesh would have required about four times as many prism cells. Attaching a rectangular plate to the lower bow increased the total resistance from 3.68 N to 4.50 N (+22 %) while the wetted surface grew by only 4 %. A flat-plate friction estimate accounts for 6 % of the increase; the remainder is attributed to pressure resistance at the plate, consistent with the local pressure rise and the distorted bow wave in the solution. The assignment provided no reference data, so each mesh decision was checked against an independent physical estimate (wave height, boundary-layer thickness, first-cell y+, friction line) instead of a target value.

Keywords: ship resistance; CFD; free surface; mesh design; wall treatment; appendage drag

1. Introduction

The accuracy of a free-surface resistance computation depends on the mesh as much as on the turbulence model. The flow around a hull combines a stagnation region and bow wave at the stem, a boundary layer that thickens towards the stern, a wake, and a Kelvin wave pattern that spreads behind the hull within a half-angle of 19.47°. Resolving all of these uniformly is unaffordable, and when the total cell count is fixed, refining one region means coarsening another. Mesh design then becomes a problem of allocating a fixed budget among physical regions.

The task was open-ended: the hull, speed and cell limit were given, and the mesh was to be made as accurate as possible, with no measured resistance to compare against. Without a reference value, a lower or higher drag cannot be called better. Each decision in this report was therefore tested for physical consistency instead: whether the free-surface band contains the computed wave, whether the predicted and observed y+ agree, how the prism stack compares with the boundary-layer thickness, and whether a force difference can be explained by an independent estimate.

This report describes such an allocation for a 1 m boat model limited to about 300,000 cells, the choice of near-wall treatment under that limit, and the use of the resulting mesh to explain the resistance penalty of a plate attached to the bow. Section 2 defines the problem, Section 3 the mesh, and Section 4 the wall treatment. Section 5 analyses the bow plate, Section 6 the sensitivity of the drag to local refinement, and Section 7 the limitations.

2. Problem definition and numerical method

The hull is 1.0 m long, 0.4 m wide and 0.2 m deep. Only the starboard half was modelled, with a symmetry plane at $y = 0$, so all forces reported by the solver are half-body values and are doubled in this report unless stated otherwise. The conditions are listed in Table 1. The simulation used Simcenter STAR-CCM+ with the

implicit unsteady solver at a time step of 0.02 s, and the drag was averaged over the last 300 time steps (44–50 s). A numerical damping zone 1 m long in front of the outlet suppresses wave reflection.

Table 1. Simulation conditions.

Item	Value
Hull length L, beam, depth	1.0 m, 0.4 m, 0.2 m
Speed U, Froude number	1.21 m/s, 0.39
Reynolds number ($\nu = 1.0\times10^{-6}$ m^2/s)	1.21×10^6
Transverse wavelength, $2\pi U^2/g$	0.94 m
Domain (x, y, z), hull centred at origin	−4…6 m, 0…3 m, −3…1.5 m
Turbulence model, free surface	k–ω SST [3], VOF
Base cell size	0.5 m
Cell limit	≈ 300,000

3. Mesh design under the cell budget

Domain. Guidelines derived from workshop cases commonly place the inlet about 1 L ahead of the bow and the outlet about 3 L behind the stern [1, 2]. Those cases, however, concern hulls much longer than the waves they generate. Here the wavelength is 0.94 m, essentially equal to L, so an inlet 1 L ahead would leave room for only one wavelength of the upstream disturbance, and an outlet 3 L behind, of which the last metre is the damping zone, would show the Kelvin pattern over only about two hull lengths. The domain was therefore kept at 3.5 L ahead and 5.5 L behind. This decision costs cells in the far field, which were saved by keeping the base size coarse.

Refinement zones. Table 2 lists the volumetric refinements. The free-surface zone uses cells that are fine in z only. Its first version extended ±0.05 m from the calm waterline, but the computed bow wave reached 0.065 m and left the fine band, so the band was widened to ±0.075 m. The wake zones are anisotropic, finer in x than in y, and extend from 1 L ahead of the bow to 3 L behind the stern. Around the hull, a box offset 22 mm from the hull surface, the estimated boundary-layer thickness (Section 4), was refined to half the hull-zone size, and contour-following bow and stern zones, defined in local coordinate systems rotated 45° to fit the stem and the transom, were refined by another factor of two. Each step between adjacent zones is therefore a factor of two in cell size. Fig. 1 shows the base mesh in the centre plane.

Table 2. Volumetric refinement zones (cell size as percentage of the 0.5 m base size).

Zone	Cell size	Purpose
Free surface, fine (±0.075 m)	Δz 1.56 % = 7.8 mm	Bow wave height 65 mm
Free surface, medium	Δz 3.13 % = 15.6 mm	Transition
Wake, fine / medium	Δx 12.5 / 25 %	Kelvin pattern, wake
Hull, fine / medium	3.13 / 6.25 %	Near-hull flow
Hull box (22 mm offset)	1.56 % = 7.8 mm	Boundary layer
Bow, stern, plate	0.78 % = 3.9 mm	Stagnation, pressure recovery

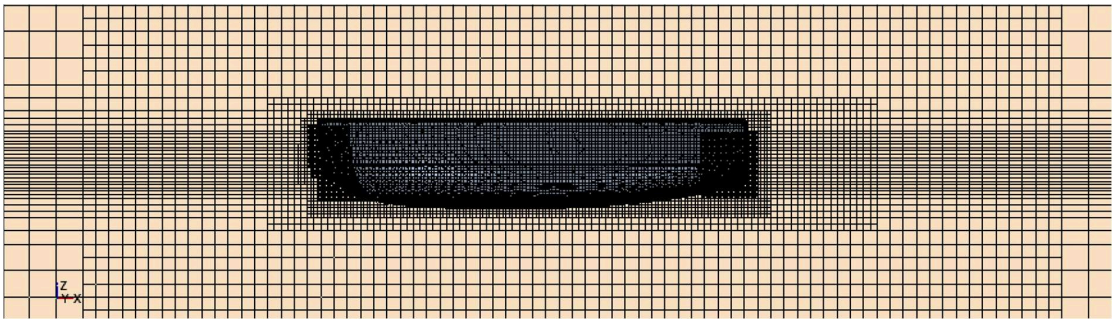


Fig. 1. Base mesh (Model A) in the x–z plane, showing the free-surface band and the hull zones; 170,732 cells.

The base mesh used 170,732 cells. The remaining budget was assigned to the hull box and the bow and stern zones, giving 298,800 cells for the bare hull. For the hull with the plate, the plate volume removed fine cells around the bow and the freed cells were used to refine the plate, giving 293,199 cells. The computed wave pattern of the bare hull is shown in Fig. 2. The diverging and transverse waves are captured up to the damping zone, but with $\Delta x = 62.5$ mm in the fine wake zone the transverse wave is resolved by only about 15 cells per wavelength and is visibly smeared further downstream.

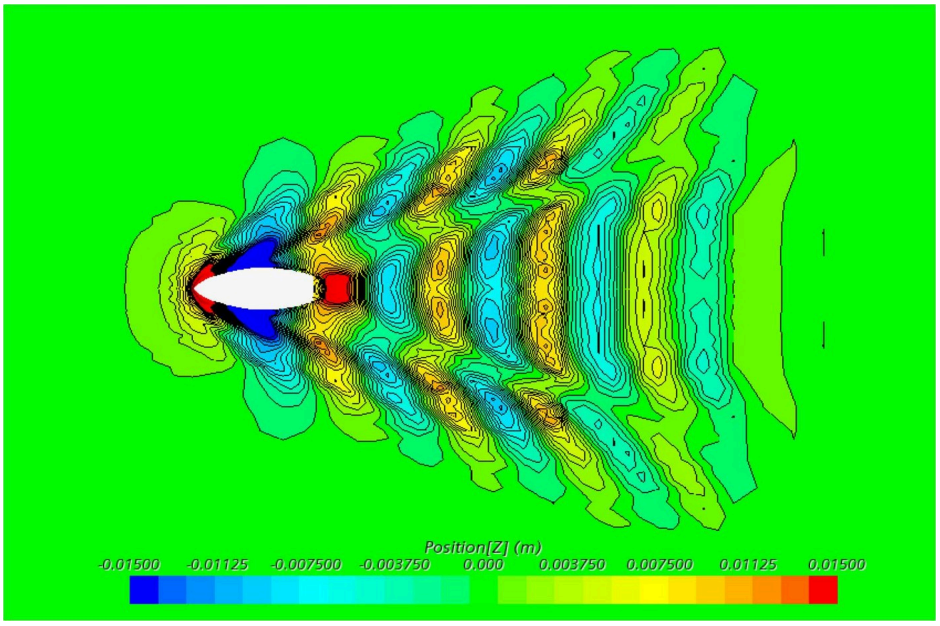


Fig. 2. Free-surface elevation around the bare hull (Model B), flow from left to right. Colour range ±15 mm.

4. Near-wall treatment

With the hull treated as a no-slip wall, the k–ω SST model is reliable either with the first cell in the viscous sublayer ($y+ \approx 1$) or with a wall function in the logarithmic region ($y+ \geq 30$); in the buffer layer, $5 < y+ < 30$, neither assumption holds. The first mesh, with five prism layers of total thickness $T = 4.26$ mm and a stretching ratio $r = 1.3$, gave $y+ = 10$–20 over the wetted hull, including the bow stagnation region. The height of the first prism layer is

$$\Delta y_1 = T\,\frac{r-1}{r^{N}-1}, \tag{1}$$

and the y+ at the first cell centroid can be estimated from flat-plate turbulent friction evaluated at mid-length, $x = L/2$ [4]:

$$y^{+} = \frac{u_\tau\,\Delta y_1/2}{\nu}, \qquad u_\tau = U\sqrt{c_f/2}, \qquad c_f = 0.0592\,Re_x^{-1/5}. \tag{2}$$

This gives $u_\tau = 0.055$ m/s. The estimates in Table 3 agree with the y+ observed in the solutions to within a factor of about 1.5, which confirms that the observed values follow from the prism-layer settings rather than

from a problem elsewhere in the mesh.

Table 3. Prism-layer settings, estimated and observed y+, and computed drag (half body).

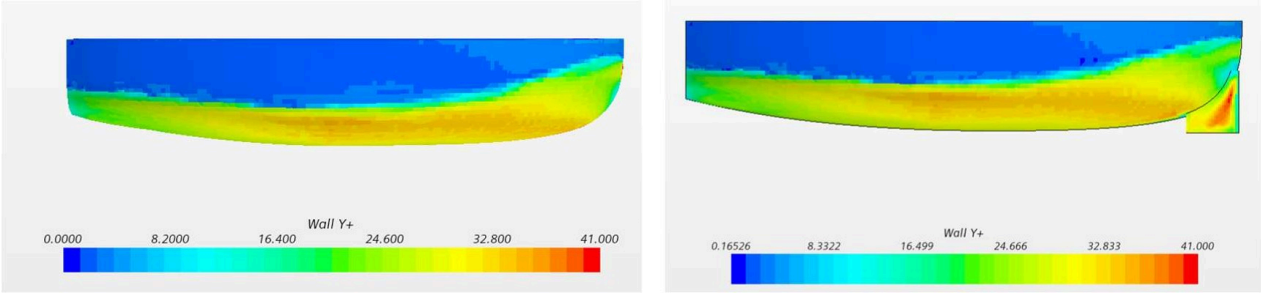
N	T (mm)	Δy_1 (mm)	y+ estimated	y+ observed	Drag (N)
5	4.26	0.47	13	10–20	1.834
12	4.26	0.057	1.6	≈ 2.5	1.817
15	4.26	0.025	0.7	≈ 1	1.786
5 (adopted)	8.0	0.88	24	≥ 30 on most of hull	1.841

Increasing the number of layers to fifteen brought y+ to about 1, and the residual of the specific-dissipation-rate equation fell from 10^{-2} to 10^{-4}. Neither result shows that the drag became more accurate. A lower residual indicates better iterative convergence, not a smaller discretisation error, and the 2.6 % fall in drag from five to fifteen layers is a change, not a convergence towards a known value. The more important issue was the thickness of the prism stack. The turbulent boundary layer at the stern is

$$\delta \approx 0.37\, x\, Re_x^{-1/5} = 0.37 \times 1.0 \times \left(1.21 \times 10^{6}\right)^{-1/5} \approx 22 \text{ mm}, \tag{3}$$

so the 4.26 mm stack covered only 19 % of it. With fifteen layers the outermost layer is 1.0 mm thick and borders hull-zone cells of 15.6 mm, a jump of fifteen times inside the boundary layer. The sublayer was resolved, but the logarithmic and outer layers were represented by one or two coarse cells. A consistent wall-resolved mesh would need the stack to span most of δ: covering 0.8δ with $\Delta y_1 = 0.025$ mm and r = 1.3 requires about 21 layers instead of 5, roughly four times as many prism cells over the whole hull, which the budget could not accommodate without removing the free-surface and wake refinements.

The wall-function approach was therefore adopted. Keeping five layers and r = 1.3 but increasing T to 8 mm (36 % of δ) raised the first-cell y+ above 30 over most of the hull (Fig. 3a). Small regions at the forefoot and the stern remain at y+ = 10–15 and are handled by the solver's blended all-y+ wall treatment. The computed half-body drag, 1.841 N, differs by less than 0.5 % from the five-layer buffer-layer case, but the near-wall model is now applied consistently over the hull.



(a) Bare hull (b) Hull with bow plate

Fig. 3. Wall y+ on the hull with the adopted prism layer (side view, bow to the right). Colour range 0–41.

5. Resistance penalty of the bow plate

A rectangular plate was attached on the centreline at the lower bow, below the waterline (Fig. 5b). The resistance results are compared in Table 4. The total resistance increased by 22 %, but the wetted surface area by only 4 %.

Table 4. Bare hull and hull with bow plate (whole body).

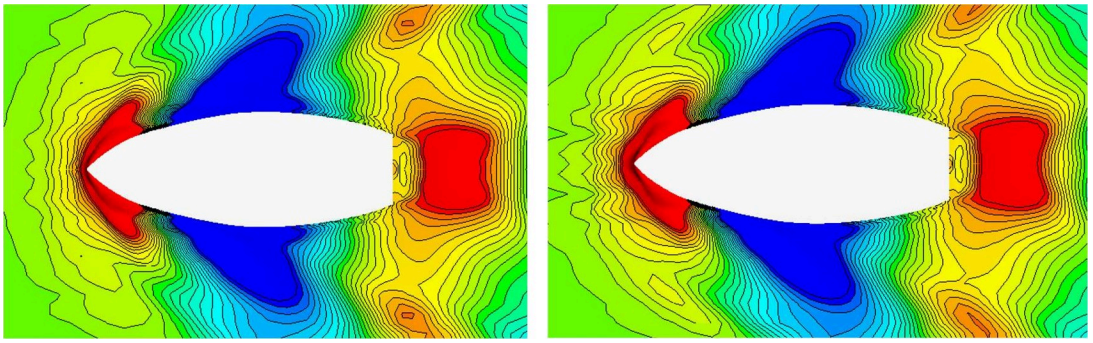
Quantity	Bare hull	With plate	Change
Total resistance R_T (N)	3.68	4.50	+22 %
Wetted surface area S (m^2)	0.375	0.390	+4 %
Friction estimate R_F, Eq. (4) (N)	1.21	1.26	+0.05 N
Number of cells	298,800	293,199	−2 %

The frictional part can be estimated with the ITTC-1957 line [5]:

$$R_F = C_F \, \tfrac{1}{2}\rho U^2 S, \qquad C_F = \frac{0.075}{\left(\log_{10} Re - 2\right)^2} = 0.00444. \tag{4}$$

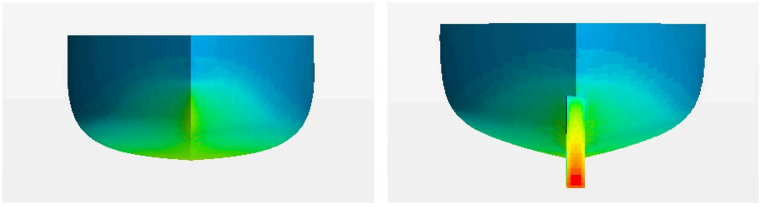
The added wetted area increases R_F by 0.05 N, which is 6 % of the observed increase of 0.82 N. The y+ distribution (Fig. 3b) remains above 30 around the plate, so the wall-shear computation is not distorted there, and there is no indication that the friction was overpredicted. About 94 % of the increase must therefore be pressure resistance.

The solution is consistent with this. The plate presents a flat face to the incoming flow and creates a second stagnation region below the stem, whereas the bare bow is a smooth curved surface with a limited stagnation area. Fig. 4 shows that the free-surface contours at the bow are distorted in the plate case while those at the stern are practically unchanged, and Fig. 5 shows a local pressure rise on the plate face with no visible difference at midship or at the stern. The pressure plotted is total pressure, which includes the hydrostatic head; part of the high value on the plate is due to its depth, but the dynamic pressure at stagnation, $\tfrac{1}{2}\rho U^2 = 730$ Pa, is a large fraction of the scale. The friction and pressure forces were not monitored separately in this study, so the split rests on Eq. (4); recording the two force components directly would make the attribution conclusive.



(a) Bare hull (b) Hull with bow plate

Fig. 4. Free-surface elevation contours near the hull, flow from left to right.



(a) Bare hull (b) Hull with bow plate

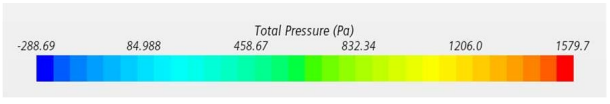


Fig. 5. Total pressure on the bow, viewed from ahead.

6. Sensitivity of the drag to local refinement

For the hull with the plate, the drag was computed with the local refinements added one at a time (Table 5). Adding the hull box, which encloses the boundary layer, reduced the drag by 3.6 %; adding the bow and stern zones reduced it by a further 0.5 %. The decreasing increments suggest that the hull-region result is becoming insensitive to further local refinement. This is not a grid-convergence study, however: the base, free-surface and wake meshes were unchanged, the refinement was not systematic, and a lower drag is not by itself a more accurate one.

Table 5. Drag of the hull with bow plate for successive local refinements (half body).

Mesh	Drag (N)	Change
Without hull box, bow and stern zones	2.344	–
With hull box	2.260	−3.6 %
With hull box, bow and stern zones	2.248	−0.5 %

7. Limitations

No model-test data exist for this hull, and a formal grid-convergence study, which requires at least three systematically refined meshes [2], was not possible within the cell limit. The numerical uncertainty of the absolute resistance is therefore unknown, and the 22 % difference between the two hulls is meaningful only because both were computed with the same mesh strategy and near-wall treatment. The wake mesh is coarse for wave propagation, so the far-field wave amplitudes are underpredicted. The full-scale extrapolation in Appendix A uses the two-dimensional ITTC-1957 method without a form factor and treats the plate's pressure drag as residuary resistance scaled by Froude similarity, which overstates its full-scale contribution if the plate drag is mainly viscous in origin.

8. Conclusions

A free-surface RANS mesh for a 1 m boat model at Fn = 0.39 was designed within a budget of 300,000 cells by keeping the domain long enough for a wavelength equal to the hull length and assigning the remaining cells to refinement zones with a factor of two between neighbours. Resolving the viscous sublayer was not compatible with the budget, because the prism stack would have needed about four times as many cells to span the boundary layer; a wall-function treatment with $y+ \geq 30$ was adopted instead. With this mesh, a bow plate increased the resistance by 22 % for a 4 % increase in wetted surface. A friction estimate accounts for only 6 % of the increase, and the local pressure rise and distorted bow wave indicate that the penalty is pressure resistance created by a second stagnation region at the plate.

References

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Appendix A. Full-scale effective power

The model results were extrapolated to a 4.2 m boat (scale ratio 4.2) by the ITTC-1957 two-dimensional method with $C_A = 0$: $C_R = C_{TM} - C_{FM}$, $C_{TS} = C_{FS} + C_R$, $P_E = R_{TS} V_S$. The model was taken in fresh water ($\rho = 997.56$ kg/m³, $\nu = 9.344 \times 10^{-7}$ m²/s, ITTC table) and the boat in sea water ($\rho = 1026$ kg/m³, $\nu = 1.189 \times 10^{-6}$ m²/s).

Table A1. Extrapolation to full scale ($V_S = 2.48$ m/s).

Quantity	Bare hull	With plate
Re_M / Re_S	1.29×10^6 / 8.76×10^6	
C_{TM}	0.01344	0.01579
C_{FM} / C_{FS}	0.00444 / 0.00307	0.00444 / 0.00307
C_R	0.00900	0.01135
S_S (m^2)	6.62	6.88
R_{TS} (N)	252	313
P_E (kW)	0.62	0.78